Title of the Project

AgriCL-Net: A Lightweight Continual-Learning System for Plant Disease Diagnosis



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Abstract

Plant diseases destroy a large share of crops every year, and checking leaves by hand is slow and often inaccurate. AgriCL-Net is an AI system that looks at a photo of a leaf and tells the farmer what disease it has, where the infection is, how bad it is, and what to do about it. The system classifies the disease, marks the infected area, measures how much of the leaf is affected, and gives a simple severity level (Mild, Moderate, or Severe). A key feature is that it can learn new diseases over time without forgetting the old ones, and it is made light enough to run on cheap edge devices like a Raspberry Pi, so it works even without a good internet connection. By combining classification, segmentation, and severity estimation in one pipeline, AgriCL-Net gives farmers, agronomists, and extension workers an objective, consistent alternative to manual inspection. The result is earlier detection, more targeted treatment, and reduced crop loss, especially for small farmers who cannot afford expensive tools.

Problem Statement

Farmers lose a large part of their crop every year to diseases that are hard to catch early. Manual inspection is slow and inconsistent, since different people judge the same disease differently, which often leads to wrong or delayed treatment. It also does not scale well to large fields, where checking every plant closely is simply not practical.

Existing AI-based solutions have their own limitations. Accurate models are usually too heavy to run on low-cost field devices like a Raspberry Pi, making them impractical for everyday farm use. They also struggle to keep up with new diseases — when retrained on a new disease, most models forget the ones they had learned earlier, a problem known as catastrophic forgetting. In addition, models trained on clean lab photos often perform poorly on real field images, which tend to have shadows, clutter, and uneven lighting.

This shows a clear need for a system that stays accurate in real field conditions, runs on affordable hardware, keeps learning new diseases without forgetting old ones, and gives farmers a simple, actionable answer.

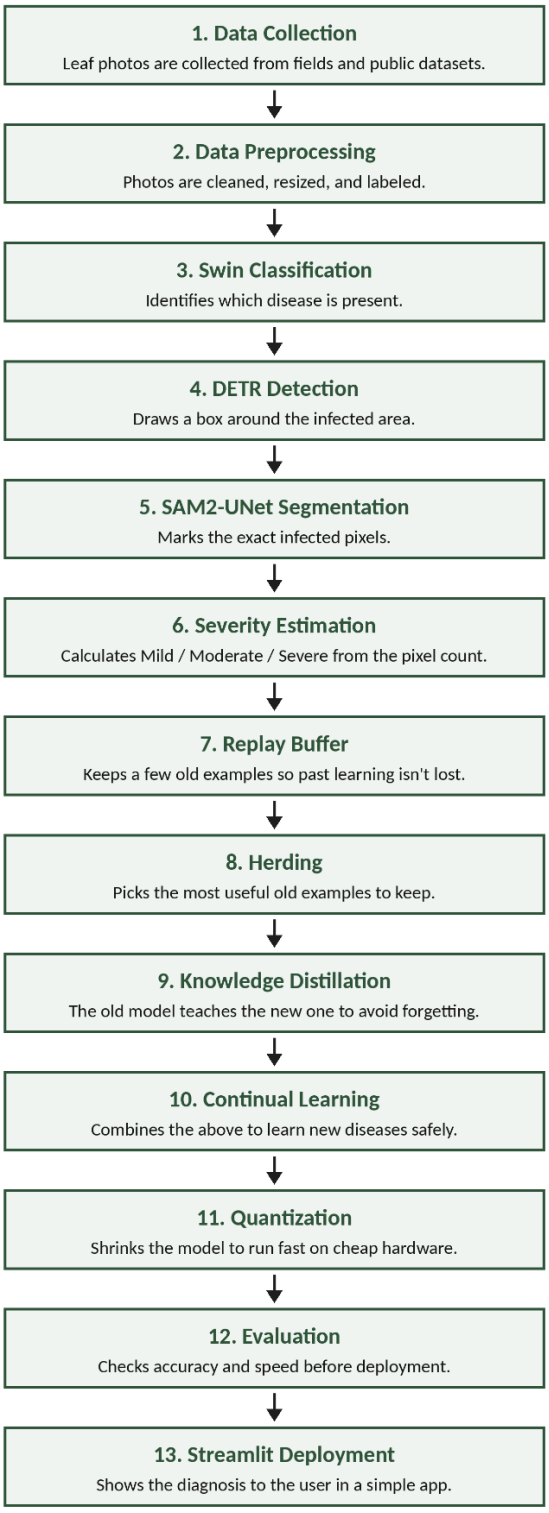
Proposed Solution

We propose AgriCL-Net, an AI pipeline that takes a leaf photo and produces a complete diagnosis: the disease name, the infected region, a pixel-level map of the damage, and a severity score. It also keeps learning new diseases after deployment without forgetting older ones, and it is compressed to run in real time on edge devices. Unlike simple classifiers, AgriCL-Net combines classification, localization, segmentation, and severity estimation in a single pipeline, so the farmer gets a complete and objective picture instead of just a disease name. Because the models are compressed and can run offline on low-cost hardware such as a Raspberry Pi or Jetson Nano, the system is practical for everyday use in the field, even where internet access is poor. The result is a tool that supports early detection, targeted treatment, and reduced crop loss.

How it works

The photo passes through a series of AI models, one after another. A Swin Transformer names the disease. A DETR model draws a box around the infected area. A SAM2-UNet model marks the exact infected pixels. From that pixel count, the system calculates a severity percentage. A separate continual-learning module (Replay Buffer, Herding, Knowledge Distillation) lets the model learn new diseases later without forgetting. Finally, the model is shrunk (quantized) and tested before being shown to the user through a simple Streamlit app.

Work FLow


1. Data Collection
Leaf photos are collected from fields and public datasets.
2. Data Preprocessing
Photos are cleaned, resized, and labeled.
3. Swin Classification
Identifies which disease is present.
4. DETR Detection
Draws a box around the infected area.
5. SAM2-UNet Segmentation
Marks the exact infected pixels.
6. Severity Estimation
Calculates Mild / Moderate / Severe from the pixel count.
7. Replay Buffer
Keeps a few old examples so past learning isn't lost.
8. Herding
Picks the most useful old examples to keep.
9. Knowledge Distillation
The old model teaches the new one to avoid forgetting.
10. Continual Learning
Combines the above to learn new diseases safely.
11. Quantization
Shrinks the model to run fast on cheap hardware.
12. Evaluation
Checks accuracy and speed before deployment.
13. Streamlit Deployment
Shows the diagnosis to the user in a simple app.

Existing Solution

A number of AI-based plant disease detection tools already exist, mostly using CNN-based image classifiers.

- **Simple disease classifiers:** Many apps can name a disease from a photo but do not show where or how bad it is.
- **Lab-only models:** Most are trained on clean lab photos and lose accuracy on real, messy field images.
- **Cloud-dependent tools:** Many need a constant internet connection and cannot run offline on cheap devices.
- **One-time trained models:** Once trained, most tools cannot learn a new disease without being fully retrained, and retraining usually erases older knowledge.

AgriCL-Net is different because it combines classification, localization, segmentation, and severity in one pipeline, keeps learning new diseases without forgetting, and is built to run offline on low-cost edge hardware.

Requirements

Functional Requirements

- Accept a leaf photo from camera or upload.
- Classify the disease present in the photo.
- Draw a bounding box around the infected area.
- Generate a pixel-level segmentation mask (background / healthy / diseased).
- Calculate a severity score and category (Mild / Moderate / Severe).
- Learn new diseases after deployment without forgetting old ones.
- Run on edge devices such as Raspberry Pi or Jetson Nano.
- Show results through a simple app interface (Streamlit).

Non-Functional Requirements

- Accuracy: diagnosis should be reliable across common diseases.
- Speed: results should appear in real time (a few seconds).
- Usability: the app should be simple enough for a farmer to use.
- Low cost: should run on affordable, widely available hardware.
- Offline capability: should not require constant internet access.
- Scalability: should support adding more diseases over time.

Hardware Requirements

- Processor: Intel Core i5 / AMD Ryzen 5 or higher
- RAM: Minimum 8 GB (16 GB recommended for smoother performance)
- Storage: Minimum 10 GB free space (SSD preferred for faster loading)
- GPU: Not mandatory to just run the app, but a GPU (e.g., GTX 1650 or better) speeds up model inference
- Webcam or camera: For capturing leaf images directly (optional if uploading images instead)
- Internet connection: Needed only for initial setup/model download; app can run offline afterward
- Operating System: Windows 10/11, Ubuntu 20.04+, or macOS
- Deployment: Raspberry Pi 4 / Jetson Nano, a camera module, and a power bank for field use.

Software Requirements

- Python, PyTorch
- OpenCV, NumPy, Pillow
- LabelMe (for annotation)
- ONNX Runtime / TensorFlow Lite (for edge deployment)
- Streamlit (for the user interface)
- Git & GitHub (for version control)

Stakeholders of the Project

The stakeholders of AgriCL-Net are the people who use the system, benefit from its features, manage the system, or are involved in developing and maintaining it. Since AgriCL-Net is designed to support the complete plant disease diagnosis process — from image capture to treatment decisions — different stakeholders have different roles and requirements.Agronomists**:** Get objective, pixel-level severity data instead of manual estimates.

1.Farmers andAgriculturalProducers

Farmers are the primary stakeholders of AgriCL-Net. They can capture or upload a photo of a leaf, receive an instant disease diagnosis, view the infected area and severity level, and get guidance on the treatment needed. This helps them act early, apply the right amount of pesticide, and reduce crop loss.

2. Agronomists and Crop Pathologists

Agronomists and crop pathologists use AgriCL-Net to get objective, pixel-level severity data instead of relying on manual visual estimation. They can verify the system's diagnosis, study disease patterns across multiple fields, and use the consistent measurements for research and advisory purposes.

3. Agricultural Extension Officers and Cooperatives

Extension officers and cooperatives can deploy AgriCL-Net across wide agricultural zones to monitor how diseases are spreading. They can use the collected data to issue timely warnings to farmers in a region and coordinate a faster response before an outbreak becomes severe

4. Agri-Tech Developers and Hardware Manufacturers

Agri-tech companies and hardware manufacturers can integrate AgriCL-Net's lightweight AI models into their own products, such as drones, farm robots, and automated spraying machines, since the models are optimized to run on low-power edge devices.

5. Government and Food Security Authorities

Government bodies and food security authorities can benefit from AgriCL-Net by using aggregated, anonymized disease data from multiple regions to plan food-security policies, allocate resources, and prepare for potential agricultural disasters.

6. System Administrator

The system administrator is responsible for managing the overall system. This includes maintaining the model updates, managing edge-device deployments, monitoring system performance, and ensuring stored data is properly maintained and secured.

7. Development Team

The development team is responsible for designing, training, testing, and maintaining AgriCL-Net. The team works on the classification, detection, and segmentation models, the continual-learning pipeline, the edge optimization, and the Streamlit-based user interface. They also improve the system based on field testing and user feedback.

8. Researchers and Educational Institutions

AgriCL-Net can also be useful for research groups, colleges, and universities. Students and researchers can use it as a base project for further work in agricultural computer vision, while institutions can use it to teach and demonstrate real-world AI applications.

How it is related to the society

- **Protects food supply:** Early detection reduces crop loss and helps food security.
- **Reduces pesticide overuse:** Precise severity data means farmers spray only where needed, which is better for the environment.
- **Helps small farmers:** Works offline on cheap devices, so it isn't limited to farmers who can afford expensive tools or internet.
- **Protects rural income:** Preventing crop loss protects the main income source for farming families.
- **Stays useful over time:** Because it keeps learning new diseases, it doesn't become outdated as new threats appear.

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